



Customer experience modeling: from customer experience to service design

Jorge Teixeira

*Madeira Interactive Technologies Institute, University of Madeira,
Funchal, Portugal and*

Faculty of Engineering, University of Porto, Porto, Portugal

Lia Patrício

Faculty of Engineering, University of Porto, Porto, Portugal

Nuno J. Nunes and Leonel Nóbrega

*Madeira Interactive Technologies Institute, University of Madeira,
Funchal, Portugal*

Raymond P. Fisk

*McCoy College of Business Administration, Texas State University,
San Marcos, Texas, USA, and*

Larry Constantine

*Madeira Interactive Technologies Institute, University of Madeira,
Funchal, Portugal*

Abstract

Purpose – Customer experience has become increasingly important for service organizations that see it as a source of sustainable competitive advantage, and for service designers, who consider it fundamental to any service design project.

Design/methodology/approach – Integrating contributions from different fields, CEM was conceptually developed to represent the different aspects of customer experience in a holistic diagrammatic representation. CEM was further developed with an application to a multimedia service. To further develop and build CEM's models, 17 customers of a multimedia service provider were interviewed and the data were analyzed using Grounded Theory methodology.

Findings – Combining multidisciplinary contributions to represent customer experience elements enables the systematization of its complex information. The application to a multimedia service highlights how CEM can facilitate the work of multidisciplinary design teams by providing more insightful inputs to service design.

Originality/value – CEM supports the holistic nature of customer experience, providing a systematic portrayal of its context and shifting the focus from single experience elements to their orchestration.

Keywords Customer experience, Service design, Interaction design, Customer service management, Design

Paper type Research paper



1. Introduction

Companies have embraced customer experience as a way to obtain sustainable competitive advantages (Shaw and Ivens, 2005), leading some authors to claim that customer experience will be the next competitive battleground (Pine and Gilmore, 1998). Meyer and Schwager (2007) define customer experience as the internal and subjective response customers have to any contact (direct or indirect) with a company. Customer experience is a holistic concept that encompasses every aspect of a company's offering (Zomerdijs and Voss, 2009).

Service design research acknowledges the importance of experience when designing a new service (Mager, 2009; Moritz, 2005). Service design is a multidisciplinary field that involves marketing, human resources, operations, organizational structure, and technology disciplines (Ostrom *et al.*, 2010). Following this holistic approach, service design orchestrates service elements such as the physical environment, people (customers and employees), and service delivery process to help customers co-create their desired experiences.

However, the scarcity of research about customer experiences (Verhoef *et al.*, 2009; Stuart and Tax, 2004; Patricio *et al.*, 2008; Roth and Menor, 2003; Hill *et al.*, 2002) is mirrored in the methods used by service design researchers to collect and depict experience data. Service design applies interdisciplinary methods and tools from several backgrounds (Moritz, 2005), but they seem focused on single elements of customer experience, rather than on the complete landscape of experience factors. For example, consider personas which are defined as "a documented set of archetypal people who are involved with a product or service" (Saffer, 2010). Personas provide information about the customer, or a specific kind of customer, who will use the service. Another example is use cases that are focused on the intended functionality of a service (Saffer, 2010). Also, another stream of service design research has focused on the service delivery process (Smith *et al.*, 2007), addressing aspects such as technology infusion (Froehle and Roth, 2004; Bitner *et al.*, 2000), customer contact intensity (Chase, 1981), and internal process design to support experience (Verma *et al.*, 1999; Voss *et al.*, 2008). Service blueprinting (Shostack, 1984; Bitner *et al.*, 2008) is one of the most common techniques for designing the service delivery process, while other operations management methods and techniques can also be applied to services (Johnston, 1999).

While these techniques focus on specific elements of customer experience, holistic study of customer experience through exploratory data collection methods will swiftly flood researchers with information. As Zomerdijs and Voss (2009) point out, it is unclear which service elements create the most compelling contexts. As such, while existing methods address some of the elements of customer experiences, there is no systematized representation of a more holistic view of the customer experience to support service design.

To fill this research gap and provide a robust toolset for service designers, we present customer experience modeling (CEM) as a method for capturing the rich and complex elements that shape an experience. CEM systematizes and represents customer experience to support service design efforts. The creative transition from understanding the customer experience to devising service solutions is crucial for service design, and models can play a key role in facilitating this transition process (Patricio and Fisk, n.d.). CEM is applied in the early stages of the service design process. CEM provides a modeling tool that enables a manageable abstraction of a complex reality and facilitates the creative transition to service design solutions. While providing a common

framework for multidisciplinary team members, CEM not only portrays the problem space, but also enables envisioning the solution space.

Several multidisciplinary contributions were included in CEM. Human activity modeling (HAM) is an interaction design tool for capturing and representing activities and their context (Constantine, 2009). HAM provides the conceptual grounding and notation for CEM. From service design and requirements engineering, we embedded in CEM the concept of customer experience requirements (CER) (Patricio *et al.*, 2008). Based on the goal-oriented analysis concept of softgoal (Mylopoulos *et al.*, 1999), CER have integrated requirements engineering and service design, bringing to the latter a way to express customer desired attributes. Employing CER's enables the assessment of how each service element influences the customer experience. Finally, multilevel service design (MSD) (Patricio *et al.*, 2011) provides the connection between customer experience and service design through three hierarchical levels that frame the experience study, from service concept to service system to specific service encounter.

In the next section, we detail how these contributions were integrated to develop CEM. Then we introduce CEM and present its application in a service design effort for a multimedia service provider.

2. From customer experience to service design

Following the service-dominant logic of Vargo and Lusch (2004) customer experience is not designed, rather it is co-created through customer interactions with the several service elements. To enable the desired experience, service designers must assemble a coherent set of elements, or clues, along the customer journey (Berry *et al.*, 2002). These service elements are the context within which an experience takes place and, along with service activities, comprise the prerequisites that service designers put forward to enable desired experiences (Zomerdijsk and Voss, 2009). Context encompasses the elements of each touchpoint, while activities unfold the experience.

Customers co-create unique experiences through their interactions with a service provider across different touchpoints, responding to the different designed elements, along with other elements that are not under an organization control, such as the social environment (Verhoef *et al.*, 2009). As such, we cannot expect to design experiences that follow predicted outcomes exactly. Instead, we only design situations that better support customers in co-creating their desired experiences (Forlizzi and Ford, 2000). While many elements take part in shaping the customer experience, it is unlikely the customer recognizes any structure behind it, instead perceiving each experience as a complex but unitary feeling (Gentile *et al.*, 2007). In this context, customer experiences cannot be designed by the organization, but services can be designed for the customer experience (Patricio *et al.*, 2011).

Existing service design methods focus on separate elements of the customer experience but designers must embrace the holistic nature of customer experience and take any and all elements and touchpoints into account (Berry *et al.*, 2002). This requires an approach that captures diverse customer experience components and systematizes them for service designers. Since some of these components are beyond a company's reach, we must examine the experience from the customer's eyes.

To that end, CEM supports the holistic nature of customer experience, provides a systematic portrayal of the experience context, and considers the physical artifacts, the technology-enabled systems, and the actors involved in each activity throughout a customer journey. Thus, we can characterize the customer experience and shift

the focus from single elements to their orchestration. CEM does not substitute for existing methods, but provides a higher-level approach that systematizes experience information to support service design early stages. To achieve a comprehensive characterization of customer experience we followed a model-based approach, which combined several multidisciplinary contributions.

3. CEM underpinnings

CEM is a model-based method that combines three multidisciplinary contributions to represent and systematize customer experiences for service design efforts. We use models to synthesize and communicate knowledge between members of a multidisciplinary service design team. Models are abstractions used to explain concepts and their relationships, which are too complex to be otherwise understood (Ludolph, 1998). Using models helps the design process by making visible a system's elements and its interactions, thus enabling the design team to achieve a common point of view, both of the problem and its solution (Dubberly *et al.*, 2008).

3.1 Human activity modeling

CEM adapts HAM (Constantine, 2009) concepts and notation to represent the rich contextual environment underlying customer experience. We also employ HAM's Participation Map, which represents several contextual elements related to an activity, such as physical artifacts, actors and systems.

HAM systematizes Activity Theory, a framework that puts activity and the tools that support it at the center of the design process. This enables us to broaden our scope of analysis and achieve a holistic representation of customer experience, considering both activities and the different contextual components that frame it.

Focusing on activities, and the context in which they are performed, also helps address a key issue when designing new services; how they meet their intended customer needs. By studying how customer's get a job done (Bettencourt, 2010), instead of merely asking their opinions on a given service, we are paving a path to better address their needs (Dahlsten, 2003). In fact, short-term customer orientation, centered on solving current satisfaction problems, may privilege incremental innovation, while damaging radical innovation (Bonner and Walker, 2004; Hillebrand, 2011; Dahlsten, 2003). As Ulwick (2002) points out, customer satisfaction studies do not provide any clue on how to achieve it, as they are unable to reveal latent customer needs (Matthing *et al.*, 2004). Other design related fields, such as interaction design, have also pointed to similar conclusions, suggesting designers take a closer look into user intentions, instead of merely supporting their current needs (Constantine, 2004). In service design, MSD (Patricio *et al.*, 2011) has already embedded the concept of activity in its approach. Interestingly, the focus on activities to design new and innovative services has permeated different fields, all suggesting a focus on customer actions as the best way to meet their needs and develop truly innovative services (Constantine, 2004; Ulwick, 2002; Bettencourt, 2010; Norman, 2005).

HAM encompasses this activity-centered perspective, but it lacks tools to evaluate the different experience related components in a way that guides service design efforts. Furthermore, as HAM hails from interaction design, it is concerned about technology systems, thus lacking the appropriate service mindset. To address this issue, we simplify HAM's approach, focusing on the most relevant and intuitive concepts

to portray customer's perspective. By limiting CEM's concepts and notation, we also make it easily understandable and practical to all stakeholders involved in the design process, especially to those least accustomed to such model-based methods.

3.2 Customer experience requirements

To effectively bridge the gap between customer experience and service design we employ two additional contributions. First, to evaluate each activity and contextual element, CEM includes CER's. CER's have their origins on requirements engineering non-functional requirements, or quality attributes, more specifically on the goal-oriented analysis softgoal concept (Mylopoulos *et al.*, 1999). As non-functional attributes are the desired qualities of a software system, they needed to be adapted to a service-oriented mindset. CER's fill this gap, being defined as the perceived attributes of the interaction with a service provider that contribute to satisfaction and usage of the service (Patrício *et al.*, 2009).

However, in CEM we apply CER's not only to a service provider but to every level of the customer experience, which may involve multiple service providers that support a certain activity. This way we are not focusing on the service provider but on the customer, thus avoiding the customer satisfaction rut mentioned before (Dahlsten, 2003). In CEM, CER's describe customer's desired qualities of an experience, thus acting as an evaluator. This way, we are able to model what customers do through the activities, how they do it through the participation map, and why they do it through CER's.

3.3 Multilevel service design

Finally, we employ MSD (Patrício *et al.*, 2011) to structure our model from an overall customer journey, to each interaction through three levels of customer experience: value constellation experience, service experience, and service encounter experience. MSD already builds upon the understanding of the customer experience to design the service offering at its different levels, and made a first integration of HAM's concepts, namely by using the activities to represent the customer experience. However, MSD representations do not address the other elements such as actors, artifacts or CERs. CEM is positioned to overcome this limitation by offering a holistic view of the elements that form the customer experience. In Figure 1 we show how each of the concepts borrowed by CEM relate to each other. This representation is recursive for each of the three MSD levels.

4. Customer experience modeling

CEM combines three multidisciplinary contributions to provide a comprehensive and systematic representation of customer experience. To apply CEM, a thorough customer study is undertaken using techniques such as interviews, observation, and contextual inquiry (Beyer and Holtzblatt, 1998).

Collected data is then systematized with three multidisciplinary contributions; HAM gives the notation and conceptual grounding for a systematic representation of the customer experience, CER's further characterize the experience by pointing out desired qualities, and the three levels of MSD structure the approach and establish the link to service design. HAM notation and concepts were adapted to reflect the customer and service focus because HAM was originally conceived to support the development of designed artifacts rather than designed services. Data must be analyzed to build the relevant categories for CEM, namely the activities performed, artifacts and systems employed, actors involved, and CER's. CEM's concepts and their notation are specified in Table I.

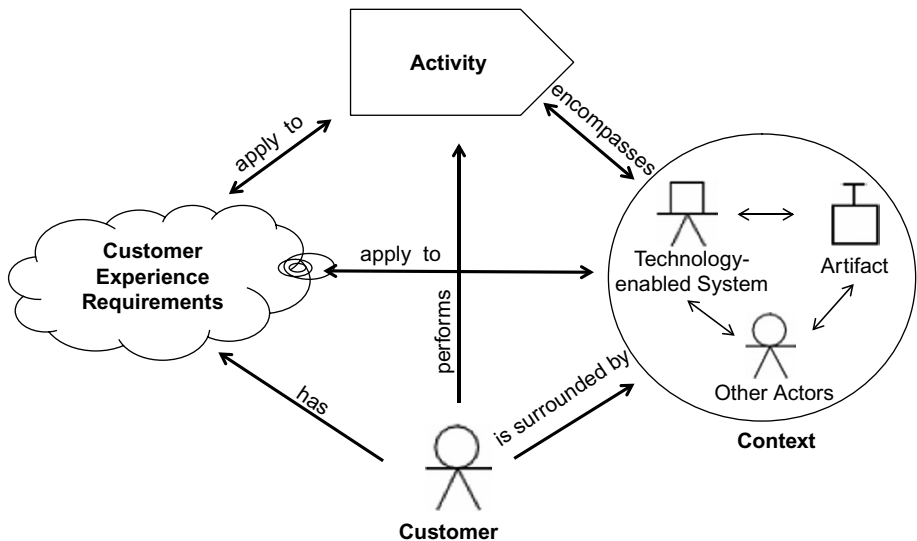


Figure 1.
CEM concepts and their
relationships

Notation	Description
 Artifact	Any artifact employed within an activity (Constantine, 2009)
 System Actor	Non-human system (software or hardware) interacting with the customer
 Actor	Activity participant interacting with the customer (or the customer himself)
 Customer Experience Requirements	Perceived attributes of the interaction with a service provider that contributes to satisfaction and usage of the service (Patricio <i>et al.</i> , 2009)
 Activity	Collection of actions or tasks undertaken for some purpose (Constantine, 2009)

Table I.
CEM notation

These are then structured according to the three levels of customer experience defined by MSD: value constellation experience, service experience, and service encounter experience. For each of them we elicit the relevant CER's and draw its participation map (on the right of Figure 1) with the contextual elements: artifacts, systems and actors.

4.1 Value constellation experience

CEM's first level, as defined by MSD, is the value constellation experience level. Value constellation experience results from interactions between the customer and all service organizations needed to perform a given customer activity. Therefore, we must first select the higher level activities, or overall activities, that apply to the specific businesses, or industries, we want to design a service for. For example, when studying the hospitality industry, we could select Accommodation as an overall activity or, for the foodservice industry, we could select Eating. Considering these broader level activities increases the possible outcomes of this kind of study. Depending on the resources available, this effort can be limited by selecting more restricted overall activities such as travel accommodation or eating at home.

Based on the selected overall activities we interview customers who perform such activities. Customers are questioned about how they perform these activities, to elicit lower level activities; what they use, or interact with when performing the activities, to elicit the contextual elements; and what they value in each activity and interacting element, to elicit CER's. Ideally, one can go from overall activities, such as Eating, to smaller actions, such as Turn on the Microwave. This enables designers to trace each decision from overall goal to single interaction.

CEM depicts experiences from the customer point of view. As such, the value constellation experience level considers not only a single service organization, but all the ones which support a given overall activity. Accordingly, the participation map depicts the contextual elements from several service providers, even if a customer does not interact directly with them. If the performance of a contextual element can have impact upon the experience, even if indirectly, CEM should reflect it.

4.2 Service experience level

After we model the relevant overall customer activities at the value constellation experience level, we focus on a single service provider service system, thus crossing to the service experience level. This level systematizes customer experience data from every service encounter with a single service provider, following the same concepts and structures from the previous level. In the service experience level, the participation map reflects the contextual elements related to the specific service provider. Following the same example as before, at this level a suitable activity for the hospitality or foodservice industry would be select a travel destination or going to a restaurant, respectively.

4.3 Service encounter experience level

The last level is the service encounter experience, where we address each specific touchpoint with the service provider. This is a very concrete level where we represent only the elements relevant to that encounter. This information could be obtained from open-ended interviews, but customers may find it difficult to recall such precise and simple activities. As such, to gather the required data for this level, it is advisable to use

contextual inquiry (Beyer and Holtzblatt, 1998), through which researchers accompany customers while they carry out the activity under consideration, and thereby capture the relevant information. Finally, examples for activities at this level would be make a reservation or ordering.

CEM aims to fill the current lack of methods by providing a systematic representation of customer experience. Combining different multidisciplinary contributions makes possible a consistent method to support service design. From interaction design, HAM provides a strong conceptual framework and a clear way to represent relevant contextual elements. From requirements engineering and service design, CER's introduce customer's desired qualities providing guidelines to service designers, while MSD structures this approach. MSD also establishes a clear-cut connection with existing design methods. Together, these contributions capture the various components of customer experience while retaining its holistic scope.

5. CEM application to a multimedia service provider

CEM was further developed and refined as part of a project to design new services and improve existing ones for a large Portuguese multimedia service provider. Its service offerings include cable TV, high-speed internet, mobile and landline phones, and several video on-demand channels. The project involved a multidisciplinary team of business specialists, designers, and software engineers, which provided a fruitful ground for CEM's application.

To begin with, we studied the service provider business model to determine the most relevant customer activities their services supported. As the huge majority of customers were final, residential consumers, and the service provider's main business goal was to maintain its market share in this segment, we excluded business customers. For residential customers, the service provider offering fit into two key overall activities; entertainment, which included cable TV and high-speed internet, and communication, which included the mobile and landline phone services. Therefore, entertainment and communication were the two overall activities we started with.

Subsequently we interviewed 17 residential customers and analyzed the collected data following a Grounded Theory approach (Corbin and Strauss, 2008; Charmaz, 2006) with the help of NVivo 8 software. This analysis provided the basis to map customer activities, contextual elements, and corresponding CER's. The latter were identified as follows:

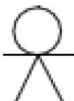
- *Affordability*. Availability at a price deemed acceptable.
- *Engagement*. The feeling of being pleasantly absorbed.
- *Content*. Availability and quantity of up-to-date multimedia materials.
- *Convenience*. Availability and ease of access to something that is desired.
- *Reliability*. Performing in a dependable way and with a predictable outcome.
- *Reward*. Worth the time spent in doing it.
- *Speed*. Swiftness in obtaining what is desired.

For each contextual element, a factsheet was produced to better describe and communicate its characteristics, mentioning relevant design guidelines and even including actual field data, such as pictures and customer's quotations. For activities, we adapted HAM's

Activity Profile (Constantine, 2009) which describes activities according to their purpose, place and time, participation, and performance characteristics. In Table II we illustrate such factsheets.

5.1 Value constellation experience

At the value constellation experience level, we selected the activity entertainment as the most relevant for this study because, as we found out, entertainment compared to communication involves a much larger range of CER's, activities, and artifacts. Such complexity provided a richer example to demonstrate CEM's capabilities. The same rationale applied for the activities chosen at the service experience level and the service encounter experience level. The example shown in Figure 2 depicts the entertainment related activities, throughout the three levels of experience, for the multimedia service provider.



Friends and Family

Characteristics: share the same space during leisure time and sometimes have conflicting interests
Design guidelines: there are conflicting interests, and designed artifacts must be tough and reliable as kids might use them
Quotation: “my son likes cartoons, my wife likes TV series [...] That’s why I have two set-top boxes, at least there are no fights!”



Set-Top Box

Characteristics: makes the interface between the customer and the company’s systems, providing added functionalities such as TV listings, recording or pay-per-view channels
Design guidelines: frequent failures seriously damage the experience as the customer wants it to be flawless and engaging
Quotation: “usually I zap between channels and I’ll see what will be on afterwards”

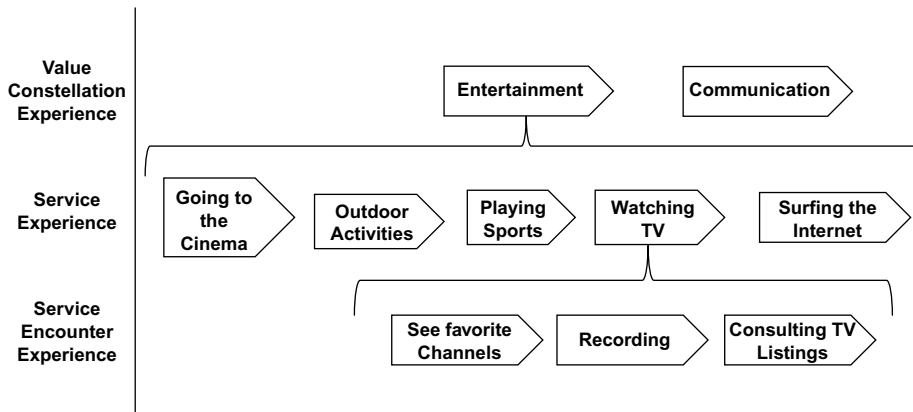


Recording

Purpose: store TV contents for later retrieval
Place and time: at home and at night, or when the customer sees something interesting enough and wants to keep it, or see it later
Participation: alone
Performance: to schedule the recording the customer has to open the TV listing and search and select the desired content. Alternatively he can start recording whenever he wants to by selecting the appropriate button on the set-top box remote

Table II.
Examples of factsheets
with descriptive data

Figure 2.
Entertainment related
activities



From the customer study, we learned that entertainment involves activities such as Playing Sports, Watching TV, and Surfing the Internet. Following the same conceptual structure shown in Figure 1, in Figure 3 we illustrate how we applied CEM throughout the three levels of customer experience, starting from the overall activity entertainment, to the activity recording.

Links between CER's and contextual elements mean that customers make an association between the two. It does not mean, however, that current elements already address those CER's, or that the CER's only apply to that element. A CER is a crosscut requirement, meaning it applies to the overall activity, and not only to a specific element. Therefore, if a CER is not linked with any contextual element it means it is a desired quality of the overall activity and, contextual elements should incorporate those qualities to better support the activity, and consequently the customer experience.

The analysis of Figure 3 reveals that entertainment is an activity rich in artifacts, such as a set-top box, high-definition (HD) and cathode ray tube (CRT) TV's, remotes for both of these, and other less technological artifacts like books and newspapers. However, none of these artifacts are associated with the CER engagement, which seems paradoxical considering the activity they support and suggests that they are viewed as a necessary means to an end. Also, reliability is always present through the different levels, suggesting

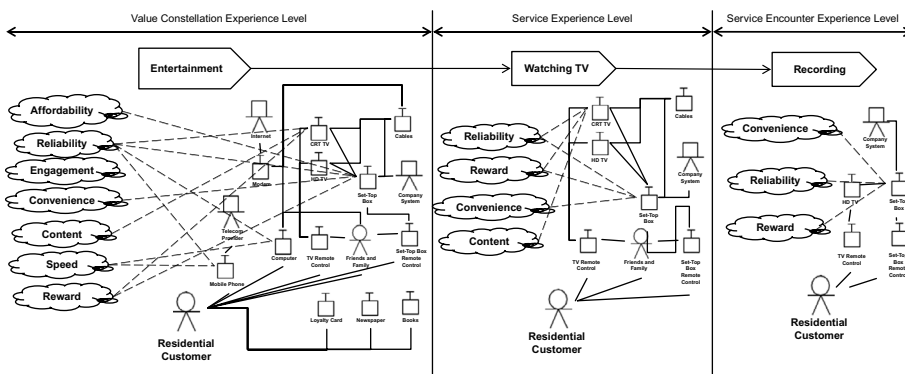


Figure 3.
CEM for a multimedia
service provider

the importance of building systems that support foolproof entertainment. This is even more important when friends and family are actors included in these activities, as is typically the case, which can imply children's involvement and the ensuing strain over such technological artifacts.

5.2 Service experience level

Following the same example shown in Figure 3, we detailed Watching TV at the service experience level, thus focusing on the multimedia service provider offering. Relevant CER's for this activity are Reliability, Reward, Content, and Convenience. Contextual elements are mostly artifacts, such as set-top box, TVs, and remotes for both. The most compelling design guidelines to be drawn from this example are concerned with the artifacts involved, with the customers even being aware of the cables that connect everything. For an activity where customers value convenience, the proliferation of artifacts is certainly not desirable. Also interesting is the association between the CER Contents and the CRT TV's, suggesting that customers with older TV sets still view them as the content providers, instead of the cable TV company.

5.3 Service encounter level

Finally, at the service encounter level we focused on the activity Recording. Reliability, Convenience, and Reward are the most relevant CER's, all of them related with the set-top box artifact. In this case, only the more technological advanced HD TVs are present, opposed to the previous level activity Watching TV, where older CRT TVs are also present. This indicates that customers perceive the difference between TV's and value them differently by performing different activities according to the type of TV they own. Viewed from the cable TV provider perspective, this can be troublesome, as a key artifact for this activity is not under company control. This CEM analysis enables envisioning possible solutions, such as bundling the set-top box and the HD TV set in a service offering.

6. Conclusion

Understanding customer experiences requires capturing rich information across all customer interactions with the service provider and even other service providers that support the overall customer activity. CEM builds upon multidisciplinary contributions in a way that systematizes this rich information and structures the holistic nature of customer experience. The application of CEM to a multimedia service shows how it can facilitate understanding the customer experience and provide valuable insights to support the service design process.

6.1 Research and managerial implications

Although customer experience has been conceptualized, empirical studies about it are scarce. There is no prior holistic approach to study and portray all the elements that form customer experience. CEM provides such an approach, contributing to support service organizations to enhance customer experience as a whole, through the integrated view of activities, actors, artifacts, and technological systems.

Service design is recognized as a human-centered approach that builds upon understanding customer experience to design service offerings. However, the richness and complexity of customer experience information makes it hard for service organizations to

analyze and incorporate it in their service design efforts. CEM's activity-centric and contextually rich representation gives service designers a comprehensive, yet manageable view of the problem space. CEM's multilevel structure also empowers service designers to trace the impact of their design decisions, from the overall service activity to each service encounter. By considering customer activities, their context, and expected qualities, CEM supplies more comprehensive representations of customer experiences than other methods. This structured, systematic view also facilitates the incorporation of customer experience into subsequent stages of service design. However, CEM is not a substitute for these other methods, but a complementary and integrative approach that provides additional inputs to service design.

Service design projects require multidisciplinary teams, interdisciplinary tools and complex methods. CEM uses concise notation to offer an intuitive visual representation of the customer experience, which can be used to facilitate communication and analysis among design teams. This enables design teams to reach a common understanding of the problem space and the solution space. Finally, CEM's multilevel structure makes the method scalable and modular which, in turn, makes its application feasible for a wide range of service design projects.

6.2 Directions for future research

This article presents the conceptual development of CEM and its application to the design of a multimedia service. This multimedia service provided a rich foundation for understanding the complexity of the customer experience and the systematic nature of CEM. New applications to other service contexts would enable further developments and refinements of the approach.

The current work should be further integrated with service design methods, especially with MSD. Forging a stronger connection between CEM and MSD would ensure the continuity of the customer focus during the whole design process. An application of CEM, across the full service design cycle, would enable further insights into how CEM models connect with service offering models to better support service design. This integration could also be taken one step further, to understand how service design decisions are interconnected with business goals and models.

Depicting and orchestrating the different elements of customer experience into different diagrams of CEM can be rather time-consuming. Developing software tools that enable the easier execution, handling, and visualization of customer experience models would significantly assist the understanding and wider application of CEM to service design projects. Such software tools would facilitate connections with models used by software engineers and interaction designers and support the work of multidisciplinary teams.

In conclusion, by combining several multidisciplinary contributions to build a model-based method, CEM captures and systematizes the holistic nature of customer experience while forging a connection with service design. The coherent models support the design process – enabling traceability from the overall service offering to each service encounter – while an established and simplified notation enhances interdisciplinary communication within and beyond the design team. Thus, service designers are able to apply CEM and infuse desired customer experiences into new or improved services.

References

- Berry, L.L., Carbone, L.P. and Haeckel, S.H. (2002), "Managing the total customer experience", *MIT Sloan Management Review*, Vol. 43, pp. 85-90.
- Bettencourt, L. (2010), *Service Innovation How to Go from Customer Needs to Breakthrough Services*, McGraw-Hill, New York, NY.
- Beyer, H. and Holtzblatt, K. (1998), *Contextual Design: Defining Customer-centered Systems*, Morgan Kaufmann, San Francisco, CA.
- Bitner, M.J., Brown, S.W. and Meuter, M.L. (2000), "Technology infusion in service encounters", *Journal of the Academy of Marketing Science*, Vol. 28, pp. 138-49.
- Bitner, M.J., Ostrom, A.L. and Morgan, F.N. (2008), "Service blueprinting: a practical technique for service innovation", *California Management Review*, Vol. 50, pp. 66-95.
- Bonner, J.M. and Walker, O.C. Jr (2004), "Selecting influential business-to-business customers in new product development: relational embeddedness and knowledge heterogeneity considerations", *Journal of Product Innovation Management*, Vol. 21, pp. 155-69.
- Charmaz, K. (2006), *Constructing Grounded Theory: A Practical Guide through Qualitative Analysis*, Sage, Thousand Oaks, CA.
- Chase, R.B. (1981), "The customer contact approach to services: theoretical bases and practical extensions", *Operations Research*, Vol. 29, pp. 698-706.
- Constantine, L. (2004), "Beyond user-centered design and user experience: designing for user performance", *Cutter IT Journal*, Vol. 17, pp. 16-25.
- Constantine, L. (2009), "Human activity modeling: toward a pragmatic integration of activity theory and usage-centered design", *Human-centered Software Engineering: Software Engineering Models, Patterns and Architectures for HCI*, Springer, London.
- Corbin, J.M. and Strauss, A.L. (2008), *Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory*, Sage, Los Angeles, CA.
- Dahlsten, F. (2003), "Avoiding the customer satisfaction rut", *MIT Sloan Management Review*, Vol. 44, pp. 73-7.
- Dubberly, H., Evenson, S. and Robinson, R. (2008), "The analysis-synthesis bridge model", *Interactions*, Vol. 15, pp. 57-61.
- Forlizzi, J. and Ford, S. (2000), "The building blocks of experience: an early framework for interaction designers", *Proceedings of Designing Interactive Systems (DIS 2000)*, ACM, New York, NY.
- Froehle, C.M. and Roth, A.V. (2004), "New measurement scales for evaluating perceptions of the technology-mediated customer service experience", *Journal of Operations Management*, Vol. 22, pp. 1-21.
- Gentile, C., Spiller, N. and Noci, G. (2007), "How to sustain the customer experience: an overview of experience components that co-create value with the customer", *European Management Journal*, Vol. 25, pp. 395-410.
- Hill, A.V., Collier, D.A., Froehle, C.M., Goodale, J.C., Metters, R.D. and Verma, R. (2002), "Research opportunities in service process design", *Journal of Operations Management*, Vol. 20, pp. 189-202.
- Hillebrand, B. (2011), "Customer orientation and future market focus in NSD", *Journal of Service Management*, Vol. 22, pp. 67-84.
- Johnston, R. (1999), "Service operations management: return to roots", *International Journal of Operations & Production Management*, Vol. 19, pp. 104-24.

-
- Ludolph, F. (1998), "Model-based user interface design: successive transformations of a task/object model", in Wood, L.E. (Ed.), *User Interface Design: Bridging the Gap from User Requirements to Design*, CRC Press, Boca Raton, FL.
- Mager, B. (2009), "Service design as an emerging field", in Miettinen, S. and Kivistö, M. (Eds), *Designing Services with Innovative Methods*, Helsinki University of Art and Design, Helsinki.
- Matthing, J., Sanden, B. and Edvardsson, B. (2004), "New service development: learning from and with customers", *International Journal of Service Industry Management*, Vol. 15, pp. 479-98.
- Meyer, C. and Schwager, A. (2007), "Understanding customer experience", *Harvard Business Review*, Vol. 85, p. 137.
- Moritz, S. (2005), *Service Design: Practical Access to an Evolving Field*, Köln International School of Design, Cologne.
- Mylopoulos, J., Chung, L. and Yu, E. (1999), "From object-oriented to goal-oriented requirements analysis", *Communications of the ACM*, Vol. 42, pp. 31-7.
- Norman, D.A. (2005), "Human-centered design considered harmful", *Interactions*, Vol. 12, pp. 14-19.
- Ostrom, A.L., Bitner, M.J., Brown, S.W., Burkhard, K.A., Goul, M., Smith-Daniels, V., Demirkan, H. and Rabinovich, E. (2010), "Moving forward and making a difference: research priorities for the science of service", *Journal of Service Research*, Vol. 13, pp. 4-36.
- Patrício, L. and Fisk, R.P. (n.d.), "Creating new services", in Fisk, R.P., Russell-Bennett, R. and Harris, L.C. (Eds), *Serving Customers: Global Services Marketing Perspectives*, Tilde University Press, Melbourne (in press).
- Patrício, L., e Cunha, J.F. and Fisk, R.P. (2009), "Requirements engineering for multi-channel services: the SEB method and its application to a multi-channel bank", *Requirements Engineering*, Vol. 14, pp. 209-27.
- Patrício, L., Fisk, R.P. and Cunha, J. (2008), "Designing multi-interface service experiences: the service experience blueprint", *Journal of Service Research*, Vol. 10, pp. 318-34.
- Patrício, L., Fisk, R.P., Cunha, J.F. and Constantine, L. (2011), "Multilevel service design: from customer value constellation to service experience blueprinting", *Journal of Service Research*, Vol. 14, pp. 180-200.
- Pine, B.J. II and Gilmore, J.H. (1998), "Welcome to the experience economy", *Harvard Business Review*, Vol. 76, pp. 97-105.
- Roth, A.V. and Menor, L.J. (2003), "Insights into service operations management: a research agenda", *Production and Operations Management*, Vol. 12, pp. 145-64.
- Saffer, D. (2010), *Designing for Interaction*, Pearson, Berkeley, CA.
- Shaw, C. and Ivens, J. (2005), *Building Great Customer Experiences*, Palgrave Macmillan, Basingstoke.
- Shostack, G.L. (1984), "Designing services that deliver", *Harvard Business Review*, Vol. 62, pp. 133-9.
- Smith, J.S., Karwan, K.R. and Markland, R.E. (2007), "A note on the growth of research in service operations management", *Production and Operations Management*, Vol. 16, pp. 780-90.
- Stuart, F.I. and Tax, S. (2004), "Toward an integrative approach to designing service experiences", *Journal of Operations Management*, Vol. 22, pp. 609-27.
- Ulwick, A.W. (2002), "Turn customer input into innovation", *Harvard Business Review*, Vol. 80, pp. 91-8.
- Vargo, S.L. and Lusch, R.F. (2004), "Evolving to a new dominant logic for marketing", *Journal of Marketing*, Vol. 68, January, pp. 1-17.

- Verhoef, P.C., Lemon, K.N., Parasuraman, A., Roggeveen, A., Tsiros, M. and Schlesinger, L.A. (2009), "Customer experience creation: determinants, dynamics and management strategies", *Journal of Retailing*, Vol. 85, pp. 31-41.
- Verma, R., Thompson, G. and Louviere, J. (1999), "Configuring service operations in accordance with customer needs and preferences", *Journal of Service Research*, Vol. 1, pp. 262-74.
- Voss, C., Roth, A. and Chase, R. (2008), "Experience, service operations strategy, and services as destinations: foundations and exploratory investigation", *Production and Operations Management*, Vol. 17, pp. 247-66.
- Zomerdijk, L. and Voss, C. (2009), "Service design for experience-centric services", *Journal of Service Research*, Vol. 13, pp. 67-82.

About the authors

Jorge Teixeira is a Doctoral Student and Teaching Assistant of the Department of Industrial Engineering and Management at the University of Porto. He holds a BSc in Economics and an MSc in Service Management and Engineering from the University of Porto. His research interests are in the field of services, with a special focus on customer experience and service design. Jorge Teixeira is the corresponding author and can be contacted at: jorgegrenha@m-iti.org

Lia Patricio is an Assistant Professor of the Department of Industrial Engineering and Management at the University of Porto. She holds a PhD in Industrial Management and Engineering, an MBA, and a Degree in Economics from the University of Porto. She has published in the *Journal of Service Research*, *Requirements Engineering Journal*, and others. Her current research focuses on new interdisciplinary methods for service system design, with a special focus on technology enabled services and complex systems.

Nuno J. Nunes is an Associate Professor of Computer Science and the President of the Madeira Interactive Technologies Institute (Madeira-ITI) at the University of Madeira. His research interests include service design, bridging software engineering (SE) and human-computer interaction (HCI), and methods and tools for agile software development. He has a PhD in HCI and SE. He's a member of the ACM and SIGCHI.

Leonel Nóbrega is an Assistant Professor at the University of Madeira and Researcher at Madeira Interactive Technologies Institute (Madeira-ITI). He holds a PhD in Software Engineering from the University of Madeira. His research focuses on modeling languages, metamodeling and model-driven development.

Raymond P. Fisk (BS, MBA, and PhD from Arizona State University) is Professor and Chair of the Department of Marketing at Texas State University. His research focuses on services. He has published in the *Journal of Marketing*, *Journal of Retailing*, *Journal of the Academy of Marketing Science*, *Journal of Service Research*, *European Journal of Marketing*, and others. He has published five books. He received the Career Contributions to the Services Discipline Award from the American Marketing Association Services Marketing Special Interest Group. He is a Member of the Distinguished Faculty of the Center for Services Leadership at Arizona State University.

Larry Constantine, IDSA, is a Professor at the University of Madeira and Institute Fellow with the Madeira Interactive Technologies Institute. He is an award-winning interaction designer specializing in safety-critical and security-related interaction design for complex systems. He is the developer of human activity modeling, and his research interests center on the role of models and modeling in systems and service design processes. He is a Fellow of the Association for Computing Machinery and recipient of the Stevens Award for his contributions to design and development methods.